

Statistical Inference: Worked Solutions

Problem 1. The empirical law and two variance estimates

Problem. For data $(0, 0, 1, 2)$, write the empirical law, its mean, its plug-in variance, and its unbiased variance estimate.

Solution. The empirical law puts mass $1/n$ on each of the $n = 4$ observations, adding the masses at repeated values. Thus

$$\hat{P}_4 = \frac{1}{2}\delta_0 + \frac{1}{4}\delta_1 + \frac{1}{4}\delta_2, \quad \bar{x} = \frac{1}{2}(0) + \frac{1}{4}(1) + \frac{1}{4}(2) = \frac{3}{4}.$$

Here δ_a denotes unit point mass at a . Its second moment is

$$\int x^2 d\hat{P}_4(x) = \frac{1}{4}(1) + \frac{1}{4}(4) = \frac{5}{4}.$$

The variance of this empirical distribution, also called the plug-in variance, is

$$V_4 = \frac{1}{4} \sum_{i=1}^4 (x_i - \bar{x})^2 = \frac{5}{4} - \left(\frac{3}{4}\right)^2 = \boxed{\frac{11}{16}}.$$

If the data are an iid sample from a population with finite variance, the unbiased estimator uses denominator $n - 1$:

$$S^2 = \frac{1}{3} \sum_{i=1}^4 (x_i - \bar{x})^2 = \frac{4}{3}V_4 = \boxed{\frac{11}{12}}.$$

These are different estimators. The empirical law itself has variance $11/16$; multiplying by $4/3$ corrects the repeated-sampling bias when estimating the population variance.

Problem 2. Uncertainty in a sample proportion

Problem. For iid Bernoulli observations with success probability $p = 0.2$, compare the variance of the sample proportion at $n = 25$ and $n = 100$. What happens to standard error?

Solution. The estimator is $\hat{p}_n = n^{-1} \sum_{i=1}^n X_i$. Because the observations are independent and each has variance $p(1 - p)$,

$$\text{Var}(\hat{p}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{p(1 - p)}{n} = \frac{0.16}{n}.$$

Hence

n	$\text{Var}(\hat{p}_n)$	$\text{se}(\hat{p}_n)$
25	0.0064	0.08
100	0.0016	0.04

Quadrupling the sample size divides variance by four and standard error by two. The variability of an individual observation remains $p(1 - p) = 0.16$; averaging changes the variability of the estimator.

Problem 3. A mean interval and a prediction interval

Problem. Assume iid Gaussian observations with unknown mean μ and known standard deviation $\sigma = 3$. For $n = 36$ and observed mean $\bar{x} = 2$, compute approximate 95% confidence and prediction intervals. State what each interval concerns.

Solution. The standardized sample mean has distribution

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

Using the rounded standard Gaussian quantile $z_{0.975} \approx 1.96$, a confidence interval for the fixed population mean is

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 2 \pm 1.96 \frac{3}{6} = 2 \pm 0.98 = [1.02, 2.98].$$

Before observing the data, this interval procedure covers the fixed μ in approximately 95% of repeated samples under the model. The exact Gaussian quantile gives exactly 95% coverage.

For an independent future observation X_{new} from the same population, the prediction error contains both observation noise and uncertainty in the estimated mean:

$$\text{Var}(X_{\text{new}} - \bar{X}) = \text{Var}(X_{\text{new}}) + \text{Var}(\bar{X}) = \sigma^2 \left(1 + \frac{1}{n}\right).$$

Its mean is zero and its distribution is Gaussian. Therefore a prediction interval is

$$\bar{x} \pm 1.96\sigma\sqrt{1 + 1/n} = 2 \pm 1.96 \cdot 3\sqrt{37/36} \approx 2 \pm 5.9611 = [-3.96, 7.96].$$

This procedure covers the future observation with approximately 95% probability over both the original sample and the future draw. Its width remains positive as $n \rightarrow \infty$, because the future observation continues to have variance σ^2 . The confidence interval for μ , by contrast, contracts as $n^{-1/2}$.

Problem 4. A beta posterior and its predictions

Problem. Begin with a Beta(1, 1) prior for a Bernoulli success probability p , and observe two successes in three conditionally independent trials. Find the posterior distribution, its mean, its interior mode, and the next-trial predictive success probability.

Solution. The prior density is uniform on $(0, 1)$. The likelihood is proportional to $p^2(1 - p)$, so Bayes' rule gives

$$\pi(p \mid x) \propto p^{1+2-1}(1 - p)^{1+(3-2)-1}.$$

Recognizing the beta density,

$$p \mid x \sim \text{Beta}(3, 2), \quad \pi(p \mid x) = 12p^2(1 - p), \quad 0 < p < 1.$$

The normalizing constant is 12 because $\int_0^1 p^2(1 - p) dp = 1/12$. The posterior mean is

$$\mathbb{E}[p \mid x] = \frac{3}{3 + 2} = \frac{3}{5}.$$

Both shape parameters exceed one, so an interior mode exists. Differentiating the log posterior on $(0, 1)$ gives

$$\frac{d}{dp} \log \pi(p \mid x) = \frac{2}{p} - \frac{1}{1 - p} = 0 \implies p_{\text{MAP}} = \frac{2}{3}.$$

The second derivative is $-2/p^2 - 1/(1 - p)^2 < 0$, verifying that this is the unique interior maximum. Posterior prediction averages the next-trial success probability over the posterior:

$$\mathbb{P}(X_{\text{new}} = 1 \mid x) = \int_0^1 p \pi(p \mid x) dp = \mathbb{E}[p \mid x] = \frac{3}{5}.$$

Thus the predictive probability uses the posterior mean, not its mode. The two summaries need not coincide even in this simple model.

Problem 5. Monte Carlo uncertainty in measured coverage

Problem. An interval procedure covers its target in 9500 of 10000 independent simulated datasets. Estimate the Monte Carlo standard error of this measured coverage. Does increasing simulation repetitions narrow each individual interval?

Solution. Let I_b indicate whether the interval constructed from simulated dataset b covers the fixed target. At the chosen generating parameter, write the true coverage as $c = \mathbb{P}(I_b = 1)$. For B independent repetitions,

$$\hat{c}_B = \frac{1}{B} \sum_{b=1}^B I_b, \quad \text{Var}(\hat{c}_B) = \frac{c(1-c)}{B}.$$

Substituting $\hat{c}_B = 0.95$ and $B = 10000$ gives

$$\widehat{\text{se}}(\hat{c}_B) = \sqrt{\frac{0.95(0.05)}{10000}} \approx \boxed{0.00218}.$$

This is about 0.218 percentage points. A normal approximation would place the simulation uncertainty for coverage at roughly $0.95 \pm 1.96(0.00218)$, or $[0.9457, 0.9543]$. This is an interval for the procedure's coverage at the simulated parameter, not an interval for the original scientific target.

Increasing B makes the measured coverage more precise. It does not change the sample size within any simulated dataset or the rule constructing each interval, so it does not narrow those intervals. Nor does calibration at one generating parameter establish the same coverage at all other parameter values.